

Your Options Trading Bot

A Beginner's Guide - no trading experience required

READ THIS BEFORE ANYTHING ELSE

This bot trades options, one of the riskiest things you can do in a stock market. With a \$500 account, a bad day can cost \$75 (15% of your money) before the bot stops itself. The bot starts in PAPER MODE - fake money, real prices. Keep it there for at least a few weeks. Nothing in this guide is financial advice. Never trade money you cannot afford to lose.

Start in 3 steps

- **1.** Double-click "Launch Options Bot.bat" in this folder. A dashboard opens in your web browser.
- **2.** Press the blue "Start" button in the left sidebar. The bot begins watching the market (it only trades 9:45 AM - 3:30 PM Eastern, Monday-Friday).
- **3.** Watch. The dashboard shows every trade, your profit/loss for the day, and the bot's reasoning in the Logs tab.

Everything below explains what the bot is actually doing and why - in plain English.

Options in plain English

A stock option is a contract that bets on where a stock's price is going, without buying the stock itself. There are only two kinds, and this bot uses both:

- **A CALL** is a bet that the stock will go UP. If you buy a call and the stock rises, your option becomes more valuable.
- **A PUT** is a bet that the stock will go DOWN. If you buy a put and the stock falls, your option becomes more valuable.

Words you'll see on the dashboard:

- **Premium:** The price you pay for the option. One contract covers 100 shares, so an option priced at \$0.40 costs \$40. This is the MOST you can lose on that trade.
- **Strike:** The stock price your bet is measured against. The bot buys strikes slightly beyond the current price (called 'out of the money') because they are cheaper.
- **Expiry:** The date the contract dies. This bot only buys options expiring in 1-7 days, and it NEVER holds one overnight - everything is sold by 3:45 PM the same day.

Why options move so fast: they are leveraged (small stock moves cause big option moves, in both directions) and they lose value every hour just from time passing ('time decay'). That is why this bot gets in and out within minutes to hours - holding is the enemy.

How the bot decides to buy

Every 30 seconds the bot checks six heavily-traded names: SPY and QQQ (funds that track the whole market) plus NVDA, TSLA, AAPL and AMZN. It is looking for a stock that just made a strong, confirmed move:

- **Momentum:** The stock moved more than 0.5% in the last 15-minute candle. Something is happening.
- **Confirmation:** An indicator called RSI confirms the move is strong (above 65 for up-moves, below 35 for down-moves). This filters out weak wiggles.
- **Crowd check:** Today's trading volume must be at least 1.5x normal. Big volume means real buyers/sellers are behind the move, not noise.

Only when ALL THREE agree does it shop for an option: up-move = buy a call, down-move = buy a put. Then the option itself must pass quality checks - it must be cheap enough (under \$50), actively traded (so the bot can sell it back instantly), fairly priced (not inflated premium), and have a tight gap between buying and selling price.

Most scans find nothing. That is normal and good - the bot is picky on purpose. Zero trades on a quiet day beats forced bad trades.

How it protects your money

Every rule below runs automatically. You do not have to do anything.

Safety rule	What it means for you
Max \$50 per trade	One contract, never more than \$50 of premium. One bad trade cannot hurt you badly.
Max 3 open trades	At most \$150 at risk at any moment.
Take profit +40%	The option is sold automatically once it gains 40%.
Stop loss -30%	The option is sold automatically once it loses 30%. Losses are cut fast.
3:45 PM flatten	Everything is sold before the close, every day, no exceptions. You never wake up to an overnight surprise.
Daily halt at -\$75	If the day's losses reach \$75, the bot sells everything, stops for the day, and sends you a Discord alert.
2-loss cool-down	Two losing trades in a row = 30 minute timeout. Stops 'revenge trading' loops.
Entry window only	No trades in the chaotic first 15 minutes or last 30 minutes of the day.

Every completed trade is written to trades.csv - what it bought, why, what it paid, what it sold for, and the result. That file is your permanent record and it is also what the AI learns from.

The AI that learns from mistakes

Each trade is saved with a snapshot of the conditions at the moment of entry: how strong the momentum was, the RSI value, the volume, the time of day, which stock, and more. After 50 completed trades, the bot trains a small machine-learning model on that history to answer one question: 'trades that looked like THIS one - did they usually win or lose?'

- **First 50 trades:** the model studies your trades silently. You will see 'learning' on the dashboard.
- **Advisory mode:** the model scores every new opportunity but cannot block anything yet. It must first prove on past data that its predictions actually beat a coin flip.
- **Gating mode:** once proven, it vetoes any trade it scores below a 45% chance of winning. The dashboard shows 'gating' with its accuracy score (AUC - above 0.55 means it has real predictive power).

In short: the strategy finds trades, the risk rules cap the damage, and the AI slowly learns which of the strategy's ideas are its worst ones - and starts skipping them.

The dashboard, piece by piece

Left sidebar

- **Mode badge:** Green 'Paper mode' badge = fake money. A red badge means LIVE real-money trading.
- **Start / Stop:** Start launches the bot; Stop closes all open trades first, then shuts down. Closing the browser tab does NOT stop the bot - only the Stop button does.
- **Settings:** Change which stocks it watches, dollar limits, profit/loss targets and the AI threshold. Press Save, then Stop and Start the bot for changes to take effect.

Main screen (refreshes every 10 seconds)

- **Daily P&L:** Today's profit or loss in dollars and as a % of your \$500.
- **Open positions tab:** What the bot is holding right now and how each position is doing.
- **Trade history tab:** Every past trade, your total results, and a chart of your account value over time.
- **Logs tab:** The bot's diary - every decision and the reason for it. If anything looks wrong, look here first.

Discord alerts (optional)

If you add a Discord webhook address to the .env file, the bot messages you on every entry, every exit, any halt, and a daily summary - so you can leave it running and stay informed from your phone.

What the files in this folder are

- **Launch Options Bot.bat** - the app you double-click.
- **trades.csv** - your trade record. Open it in Excel anytime.
- **.env** - your private keys and settings. NEVER share this file or its contents with anyone.
- **model.pkl** - the AI's saved brain and its training settings.
- **The .py files** - the bot's code: scanner (finds moves), options_chain (picks contracts), executor (buys/sells), risk_manager (the safety rules), learner (the AI), alerts (Discord), app (the dashboard), main (ties it all together).
- **settings.json / status.json / bot.log** - settings saved from the dashboard, the bot's status heartbeat, and log files. Safe to ignore.

Before you even think about real money

- Run paper mode for AT LEAST 3-4 weeks. This also gives the IV filter and the AI time to warm up - both need history to work.
- Check trades.csv weekly. Is it actually profitable AFTER a real stretch of days? A lucky week proves nothing.
- Understand that paper results are optimistic: real orders fill slightly worse than paper ones.
- Going live requires TWO deliberate changes in .env: setting LIVE_MODE=true AND swapping in your live Alpaca keys. That friction is intentional.
- If live trading starts badly, stop. The bot halting at -\$75 daily is a circuit breaker, not a strategy.

THE HONEST TRUTH

Most short-term options traders lose money. The edge this bot hunts for is small and may not exist in every market. Its real strengths are discipline - it never panics, never doubles down, always takes the stop loss - and a paper trail you can study. Treat the first months as tuition: the goal is to learn whether the strategy works, not to get rich.

Mini glossary

- **Alpaca:** The brokerage whose service the bot trades through.
- **Paper trading:** Simulated trading: real market prices, fake money.
- **Candle:** A bar summarizing the price action of one time slice (here, 15 minutes).
- **RSI:** Relative Strength Index - a 0-100 score of how strong a recent move is.
- **Bid / Ask:** The price buyers will pay / sellers want. The bot needs these close together.
- **Open interest:** How many contracts exist. High = easy to buy and sell quickly.
- **IV (implied volatility):** How expensive an option is relative to expected movement. The bot avoids overpriced ones.
- **P&L:** Profit and loss.
- **Flatten:** Sell everything, hold nothing.